both *over-engineering* (expensive sensors generating data you can't use) and *under-engineering* (missing important events because you sampled too slowly). Table 3.1 shows examples of when higher performance sensors might be required.

Table 3.1 A comparison of situations where higher performance is required across the dimensions of accuracy, sampling rate, and latency, and when more moderate performance will do.

Dimension	High performance required	Moderate performance sufficient
Accuracy	**Leak detection: ±0.01 L/min**. Small flow changes indicate leaks; false positives waste investigation time, false negatives allow damage	**HVAC control: ±0.5°C** System can't control more precisely; ±0.1°C accuracy would waste money on unused precision
Sampling rate	**Vibration monitoring: 10,000 Hz** bearing defects create high-frequency signatures; slower sampling misses critical failure indicators	**Energy monitoring: 1 sample/hour** Sufficient to detect when equipment runs and calculate daily usage; faster sampling drains batteries without enabling better decisions
Latency	**Pressure relief: <1 second** Safety-critical response to prevent equipment damage or injury requires immediate action	**Soil moisture: minutes to hours** Changes occur slowly; irrigation decisions tolerate significant delays without impact

3.2.4 Evaluate practical constraints

Real-world constraints often dictate sensor choices more than technical specifications, with power availability usually being the primary constraint. Mains-powered sensors can transmit continuously over WiFi, while battery-powered locations demand low-power wireless protocols and infrequent transmissions. Most industrial sites require certified electricians for any powered installation, adding hundreds of dollars of labor cost to each sensor. Maintenance access is equally important when considering battery-powered devices. A sensor located somewhere difficult to access needs a multi year battery life to minimize maintenance costs. My home digital twin takes a typical approach of investing in quality sensors for important measurements (power monitoring) while using $20 consumer sensors for less essential data